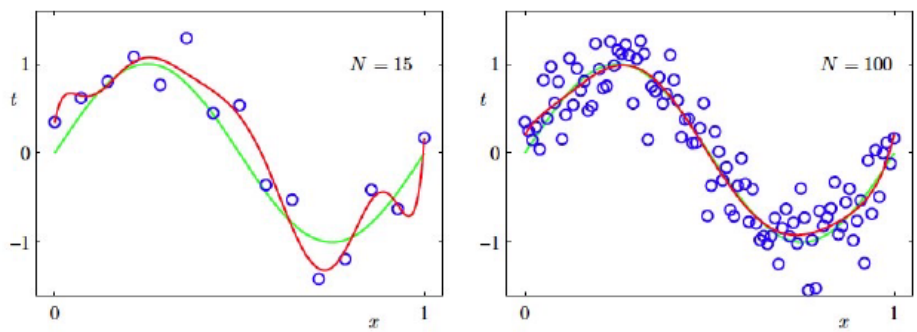
Probabilistic Machine Learning

Probability Theory

1. The "Why" of Probabilistic ML

We are looking at how algorithms handle the inherent chaos and noise of the real world. Because knowing exactly how confident your model is becomes just as important as the prediction itself

We start with a fundamental intuition: **you cannot base a decision on a prediction without quantifying the uncertainty of that prediction**.



If you look at the graphs on , you'll see regression models trying to fit data points. The models have two types of uncertainty to deal with:

- **Aleatoric Uncertainty:** This is the inherent noise or non-determinism in the data itself. Think of measurement errors, corrupted sensor data, or simply the natural randomness of tossing a coin. You cannot eliminate this uncertainty, even with infinite data.
- **Epistemic Uncertainty:** This comes from our own lack of knowledge or the finite size of our training data. Notice how the red and green curves on Slide 4 diverge significantly where there are fewer data points (N=15) compared to when there is more data (N=100). You *can* reduce epistemic uncertainty by gathering more data.

2. The Core Framework: Discrete Random Variables

Probability theory gives us the framework to define and manipulate this uncertainty mathematically. Instead of deterministic outputs, we treat outcomes as **Random Variables (RVs)**, which measure specific aspects of an experiment's outcome. We can quantify the uncertainty about learned model parameters and predictions

A Random Variable X, is a variable where its takes a value from a set of possible values x (called the domain, D_x), where each possible value comes with a probability.

- For example, X takes the values x = {1,2,3} where it has a probability of 0.2 of take 1, 0.5 of taking 2 and 0.3 of taking 3
- it is a Mathematical model of (repeatable) random experiments

Abstract notation

- ❑ Random variable X (capital letter)
- ❑ Domain D_X
- ❑ Concrete value $x \in D_X$ (non-capital letter)
- ❑ Event is subset $\mathcal{X} \subseteq D_X$ (e.g., even dice sum: $\mathcal{X} = \{2, 4, \ldots, 10, 12\}$)

Probability distributions

In limit of total number of trials (n) to infinity, can define the probability of an event as limit of fraction:

$$\mathrm{P}(X \in \mathcal{X}) = \lim_{n \to \infty} \frac{\#\text{trials with } X \in \mathcal{X}}{n}$$

Set of all those probabilities is called the probability distribution of X.

For a **Discrete Random Variable**, the domain is a set of distinct possible values (like heads/tails, or dice sums from 2 to 12).

RV is called discrete if there is probability mass function $p_X(x): D_X \to \mathbb{R}_+$ such that for all events $\mathcal{X} \subseteq D_X$

$$\mathrm{P}(X \in \mathcal{X}) = \sum_{x \in \mathcal{X}} p_X(x)$$

Example: sum of two dice

X	p
2	1/36
3	2/36
4	3/36
5	4/36
6	5/36
7	6/36
8	5/36
9	4/36
10	3/36
11	2/36
12	1/36

$$\begin{aligned}\mathrm{P}(X \in \{2,4,\dots,12\}) &= p_X(2) + p_X(4) + \cdots + p_X(12) \\ &= \frac{1}{36} + \frac{3}{36} + \frac{5}{36} + \frac{5}{36} + \frac{3}{36} + \frac{1}{36} \\ &= \frac{18}{36} = \frac{1}{2}\end{aligned}$$

Probabilistic Machine Learning

- **Probability Mass Function (PMF):** This function, $p_X(x)$, maps each possible outcome to a probability.
- **The Golden Rules:** Probabilities must be bounded between 0 and 1, and the sum of all possible probabilities in the domain must exactly equal 1 ($\sum_{x \in D_X} p_X(x) = 1$).

3. Multiple Variables: Joint, Marginal, and Conditional

Real-world ML problems rarely involve just one variable. Let's look at how multiple variables interact using the double coin toss example:

Define three random variables

X: tossing the coin for the first time ($D_X = \{H, T\}$)
Y: tossing the coin for second time ($D_Y = \{H, T\}$)
Z: number of head outcomes ($D_Z = \{0, 1, 2\}$)

Joint probability (mass function) $p(\cdot,\cdot,\cdot)$ gives probability of possible combination of values:

X	Y	Z	p
T	T	0	4/9
T	H	1	2/9
H	T	1	2/9
H	H	2	1/9

(omitted entries w. $p = 0$)

marginal probability

$$p_Z(z) = \sum_{x,y} p(x, y, z)$$

Z	p
0	4/9
1	4/9
2	1/9

Again, all probabilities sum to 1: $\sum_{x,y,z} p(x, y, z) = 1$

- **Joint Probability:** The probability of a specific combination of events happening together, denoted as $p(x, y, z)$.
- **Marginal Probability:** This is the probability of one variable ignoring the others.
- "Marginalizing" simply means collapsing a dimension. If you have a 2D table of probabilities for X and Y, you find the marginal probability of X by summing up all the rows for Y.

Double Coin Toss Example

X: tossing the coin for the first time ($D_X = \{H, T\}$)
Y: tossing the coin for second time ($D_Y = \{H, T\}$)
Z: number of head outcomes ($D_Z = \{0, 1, 2\}$)

What is the probability of two heads in total given that the first toss comes up heads?

$$p_{Z|X}(2 \mid H) = \frac{p_{Z,X}(2, H)}{p_X(H)} = \frac{1}{3}$$

- **Conditional Probability:** The probability of $X = x$ happening *given* that we know $Y = y$ has happened:
  - $p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$.
- Product rule (chain rule)
  - $p\ (x, y) = p\ (x \mid y)\ p(y)$
- Bayes' Theorem
  - $p(y|x) = \frac{p(x|y)p(y)}{p(x)}$

Example task: compute probability of first coin "heads" given that total of one "head"

$$p_{X|Z}(H|1) = \frac{p_{Z|X}(1 \mid H)p_X(H)}{p_Z(1)} = \frac{\frac{2}{3}\cdot\frac{1}{3}}{\frac{4}{9}} = \frac{1}{2}$$

- **Independence:** Two variables are independent if the occurrence of one doesn't affect the other.
  - Mathematically, the joint probability is just the product of their individual probabilities:
  - $p_{X,Y}(x, y) = p_X(x)p_Y(y)$.
  - $p(x) = p(x|y)$

Double Coin Toss Example

X: tossing the coin for the first time ($D_X = \{H, T\}$)
Y: tossing the coin for second time ($D_Y = \{H, T\}$)
Z: number of head outcomes ($D_Z = \{0, 1, 2\}$)

X and Y are independent

X and Z are dependent

$$0 = p_{X,Z}(H, 0) \neq p_X(H)p_Z(0) = 4/27$$

4. Continuous Variables and Expectations

When a variable can take on any real number (like time or distance), we use **Continuous Random Variables**.

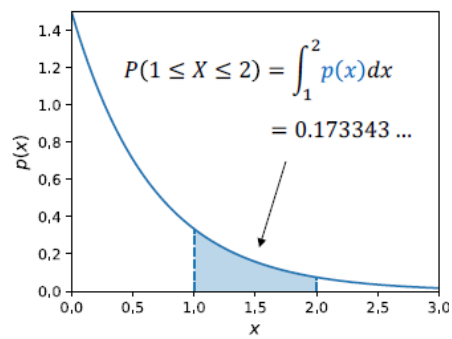
- Continuous random variables X with domain $D_x \subseteq \mathbb{R}$ and $P(X = x) = 0$
- Instead of a PMF, we use a **Probability Density Function (PDF)**, $p(x) \geq 0$ that describes probability of event $a \leq X \leq b$ by integration
- Because there are infinite possible specific points, the probability of hitting one exact infinitely precise point is zero ($P(X = x) = 0$).
- Instead, we find the probability of a value falling within a *range* by taking the integral:
  - $P(a \leq X \leq b) = \int_a^b p(x)dx.$

Example: time until you receive first email on random workday

$$p(x) = \frac{3}{2}\exp\left(-\frac{3}{2}x\right)$$

Question: probability of receiving first mail in second hour of work?

Probabilistic Machine Learning



- **Crucial Trap :** Densities are *not* probabilities. A density function can actually have a value greater than 1! The only strict rule is that the total area under the curve (the integral from $-\infty$ to ∞) must equal exactly 1.

Expected Value ($E[X]$) is essentially the "center of mass" or the long-run average of the random variable.

You calculate it by multiplying every possible outcome by its probability (or density) and summing (or integrating) them all up.

If $D_X \subseteq \mathbb{R}$, we define expected value $\mathbf{E}[X]$ of X as:

$$\mathbf{E}[X] = \int_{D_X} x\, p(x)\, dx \qquad \mathbf{E}[X] = \sum_{x \in D_x} x\, p(x)$$

continuous X discrete X

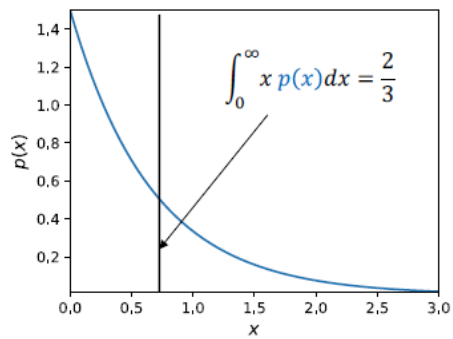
- For Continuous RV

Question: when do I receive my first email "on average"

$$p(x) = \frac{3}{2}\exp\left(-\frac{3}{2}x\right)$$

See Activity 0.2 for more details

Probabilistic Machine Learning



- for Discrete RV

Question: What is the average number of "head" tosses

$$\mathbf{E}[Z] = 0 \cdot \frac{4}{9} + 1 \cdot \frac{4}{9} + 2 \cdot \frac{1}{9} = \frac{2}{3}$$

Z	p
0	4/9
1	4/9
2	1/9

Conditional Expectations, $E[X|y = y]$

For $y \in D_Y$ with $p(y) > 0$, we can also define conditional expectation:

$$\mathbf{E}[X \mid Y = y] = \int_{D_X} x\, p(x \mid y)\, dx \qquad \mathbf{E}[X \mid Y = y] = \sum_{x \in D_X} x\, p(x \mid y)$$

continuous X discrete X

Question: what is the expected number of "head" outcomes given that the first toss is "tail" / "head":

$$\mathbf{E}[Z \mid X = T] = 0 \cdot \frac{2}{3} + 1 \cdot \frac{1}{3} + 2 \cdot 0 = \frac{1}{3}$$

$$\mathbf{E}[Z \mid X = H] = 0 \cdot 0 + 1 \cdot \frac{2}{3} + 2 \cdot \frac{1}{3} = \frac{4}{3}$$

X	Y	Z	p
T	T	0	4/9
T	H	1	2/9
H	T	1	2/9
H	H	2	1/9

Uncertainty

1. From Predictive to Probabilistic Models

Up until now, you have likely built **predictive models**. You feed in some features (x) and weights (w), and the model spits out a deterministic prediction, $y(x, w)$.

A **probabilistic model** changes the paradigm. Instead of just giving you a target y, it gives you a conditional probability distribution: $p(y|x, w)$.

- **Insight:** Why do we care? Because in the real world, being confidently wrong is dangerous. A probabilistic model doesn't just say "The patient has this disease"; it says "Given the symptoms, there is a 92% probability of this disease." It provides both a prediction and a quantifiable level of certainty.

The Setup: Imagine we flip a single specific coin, A, 10 times and get this dataset: $D = \{H, T, T, T, T, H, T, T, H, T\}$.

- We have 3 Heads and 7 Tails.
- We want to find a parameter w (the probability of getting a Head) that best models this specific coin A.
- build a model for this coin tossing to estimate the probability that next coin toss will be a head (or tail).

Our **model** has one parameter $w \in [0, 1]$ that models probability of head:

$$p(y|w) = \begin{cases} w, & y = H \\ 1 - w, & y = T \end{cases}$$

Problem: need to estimate w from training data D.

2. Maximum Likelihood Estimation (MLE)

Maximum Likelihood Estimation is one of the most important concepts in all of statistics.

The core idea: The prediction problem (probability that the next coin is H or T) is formulated as finding an optimal coin toss model (functions of parameter w) that maximizes the probability of observing the data.

We want to maximize the likelihood function, $p(D|w)$.

Simpler English: **Out of all the possible coins in the universe, which one is most likely to have generated the exact sequence of data we just saw?**

maximiser w_{ML} is called maximum likelihood estimate:

- $p(D|w_{ML}) = max_w p(D|w)$

With independent data D = {y+,..., yn} likelihood becomes product

$$p(D|w) = p(y_1|w)p(y_2|w) \dots p(y_n|w) = \prod_{i=1}^{n} p(y_i|w)$$

Because each coin toss is an independent event, the total probability of seeing our exact dataset is just the product of the individual probabilities:

- $p(D|w) = w^{|H|}(1-w)^{|T|}$

For our specific dataset, this is $w^3(1-w)^7$.

The "Log Trick" : If we want to find the maximum of a function, we take the derivative and set it to zero ($\frac{\partial}{\partial w}p(D|w) = 0$). However, taking the derivative of a massive chain of multiplied probabilities is a mathematical nightmare.

❑ **D = {H, T, T, T, T, H, T, T, H, T}**

$$p(D|w) = w^3(1-w)^7$$

If w had been either 0 (heads impossible) or 1 (tails impossible), then the probability of getting D would have been 0.

If w had been ½ (a fair coin), then the probability of getting D would have been 1/1024.

- **Insight:** There is also a computational reason for this! If you multiply thousands of probabilities (numbers between 0 and 1) together in Python or C++, your computer will suffer from "floating-point underflow" and just round the answer to 0.

To fix both the math and the code, we take the **natural logarithm** of the likelihood function. Logarithms turn multiplication into addition ($ln(a \cdot b) = ln(a) + ln(b)$). Because the logarithm is a strictly increasing function, the w that maximizes the log-likelihood is the exact same w that maximizes the regular likelihood!

Idea: use derivative criterion to identify max of likelihood function

$$\frac{\partial}{\partial w} p(D|w)\Big|_{w=w_{\mathrm{ML}}} = 0$$

Problem: product very hard to differentiate

Instead: consider log likelihood function (has same maximiser)

$$\ln p(D|w) = \ln \prod_{i=1}^{n} p(y_i|w) = \sum_{i=1}^{n} \ln p(y_i|w)$$

and work with

$$\frac{\partial}{\partial w} \ln p(D|w) = \sum_{i=1}^{n} \frac{\partial}{\partial w} \ln p(y_i|w)$$

Data

$$D = \{H, T, T, T, T, H, T, T, H, T\}$$

Likelihood

$$p(D|w) = w^3(1-w)^7$$

Log likelihood

$$\ln p(D|w) = 3 \ln w + 7 \ln(1-w)$$

Derivative criterion

$$\frac{\partial}{\partial w} \ln p(D|w) = \frac{3}{w} - \frac{7}{1-w} = 0$$

Result: $w_{\mathrm{ML}} = 0.3$

Likelihood

$$p(D|w) = w^{|H|}(1-w)^{|T|}$$

Log likelihood

$$\ln p(D|w) = |H| \ln w + |T| \ln(1-w)$$

Derivative criterion

$$\frac{\partial}{\partial w} \ln p(D|w) = \frac{|H|}{w} - \frac{|T|}{1-w} = 0$$

Maximum likelihood estimate

$$w_{\mathrm{ML}} = \frac{|H|}{|H| + |T|}$$

When we take the derivative of the log-likelihood and solve for zero, we get a beautifully simple result:

$w_{ML} = \frac{|H|}{|H|+|T|}$

For our data, $w_{ML} = \frac{3}{10} = 0.3$. The math perfectly matches our intuition!

---

3. Uncertainty and The Bootstrap

MLE is great, but it has a glaring flaw: it gives us a **point estimate** (a single number, 0.3).

Remember our types of uncertainty?

- **Aleatoric:** The inherent non-determinism of modelled process
  Example: uncertainty of outcome of coin flip even if exact probability known
- **Epistemic:** Our lack of knowledge due to the finitude of our specific observed dataset

  Example: because we only flipped the coin 10 times.

Assume we had a specific (random) dataset D to find parameters $\boldsymbol{w}(D)$, e.g., with maximum likelihood.

- Observation: that makes $\boldsymbol{w}$ as well as individual parameters w a random variable with some distribution
- Conceptually: the less the distribution of w is concentrated, the more uncertain we are about our specific estimate $w(D)$
- Idea: if we could sample new datasets, we could start to estimate that distribution (and the dispersion / variance of w)
- Problem: in practice we have only one dataset, and we would like to use all data for learning w

If I flipped a coin 10 times and got 3 heads, MLE says w = 0.3. If I flipped it 10,000 times and got 3,000 heads, MLE *still* says w = 0.3. But clearly, I am vastly more confident in the second estimate! How do we mathematically capture this epistemic uncertainty if we only have one dataset?

The Solution: Bootstrapping
Bootstrapping is a brilliant frequentist hack. Since we can't go out and generate infinite new datasets, we simulate it by resampling our *existing* dataset **with replacement**.

1. Put all 10 of your original coin tosses (original data) into a bag.
2. Draw one out (an observation of the original data), record it, and **put it back**.
3. Repeat this 1-3N times to create a "bootstrap sample", where N is the number of observations in the original sample
4. We will end up with a bootstrap sample with 1-3N observations
5. Calculate w_{ML} for this new sample.

Because we sample *with replacement*, some tosses will be picked multiple times, and some won't be picked at all. If you do this thousands of times, you will get a distribution of w_{ML} values. The spread (variance) of this distribution tells you exactly how uncertain your model is about its parameters!

We are shifting from the Frequentist worldview (where parameters are fixed but unknown) to the **Bayesian Approach** (where we treat the parameters themselves as random variables).

Bayesian Approach

- use probabilities to quantify the uncertainty directly

1. The Flaw of Maximum Likelihood

We left off knowing that Maximum Likelihood Estimation (MLE) looks purely at the data. But what happens when you have almost no data?

Suppose you flip a coin twice, and you get Tails both times ($D = \{T, T\}$). If you plug this into our MLE formula:

$w_{MLE} = \frac{H}{H+T} = \frac{0}{0+2} = 0.$

MLE confidently tells you that this coin has a 0% chance of ever landing on Heads. This is obviously unreasonable. From your lived experience, you know that almost all coins are roughly 50-50.

We need a mathematical way to encode this prior experience, and that is exactly what the Bayesian approach does.

2. Bayes' Theorem and The Beta Distribution

The Bayesian approach asks us to define a **Prior Distribution**, $p(w)$, which captures our beliefs about the parameter *before* we see any data. We then update this belief using the data via Bayes' Theorem to get the **Posterior Distribution**, $p(w|D)$.

$p(w|D) = \frac{p(D|w)p(w)}{p(D)}$

- $p(D|w)$ is the Likelihood (our MLE function).
- $p(w)$ is the Prior belief - probabiliy distribution over parameter w
- $p(D)$ is the marginal probability of the data (mostly acting as a normalizing constant here). - probability of observed data D

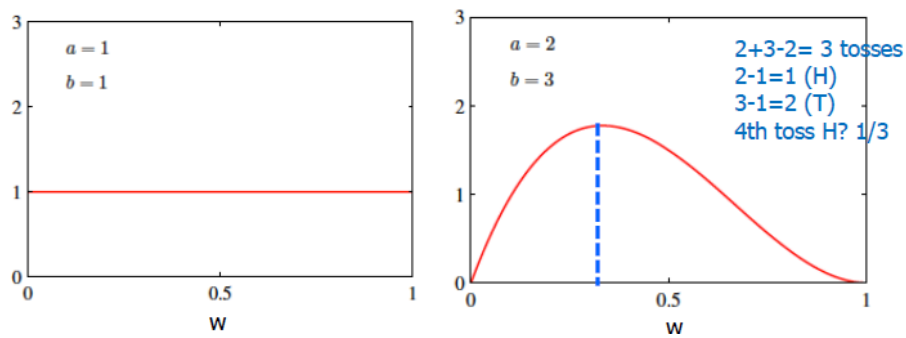
The Beta Prior:

To model our prior belief about a probability w (which must live between 0 and 1), we use the **Beta distribution**:

$p(w) \propto Beta(w|a, b) \propto w^{a-1}(1-w)^{b-1}$, w = [0, 1].

The parameters a and b are brilliant: you can think of $(a-1)$ and $(b-1)$ as "pseudo-counts" or imaginary coin tosses you've observed in your past experience.

How to interprete: (a-1) and (b-1) are the number of Heads and Tails I think we would see, if we made (a+b-2) coin tosses!



- If you set $a = 100$ and $b = 100$, you are essentially saying, "I have a very strong prior belief this coin is fair, as if I've already seen 99 heads and 99 tails".
- this is similar to the MLE of $w^3(1-w)^7$

Bayes' Theorem:

$$p(w|D) = \frac{p(D|w)p(w)}{p(D)}$$

Let's put together the numerator!

$$p(D|w) = w^{|H|}(1-w)^{|T|} \text{ (by MLE)}$$
$$p(w) \propto w^{a-1}(1-w)^{b-1} \text{ (by prior belief)}$$

So we get the numerator:

$$p(w|D) \propto w^{|H|+a-1}(1-w)^{|T|+b-1}$$
$$\propto \text{Beta}(w||H|+a, |T|+b)$$

Updating to the Posterior: When we multiply the MLE ($p(D|w) = w^{|H|}(1-w)^{|T|}$) by our Beta Prior belief ($p(w) \propto w^{a-1}(1-w)^{b-1}$), the exponents simply add

together.
The Posterior becomes: $Beta(w||H|+a,|T|+b)$.

You are literally just adding your real data observations ($|H|$ and $|T|$) to your prior imaginary observations (a and b).

3. Bayesian Prediction

Now, how do we make a prediction? With MLE, we just plugged in the single best w_{ML}. In Bayesian ML, we acknowledge our uncertainty. We don't just use one w; we integrate over *every possible value* of w, weighted by how likely that w is according to our posterior.

$p(H|D) = \int_0^1 p(H|w,D)p(w|D)dw = E[w|D]$

Probability of head given data

$$
\begin{aligned}
p(H|D) &= \int_0^1 p(H,w|D)dw && \text{sum rule} \\
&= \int_0^1 p(H|w,D)p(w|D)dw && \text{chain rule} \\
&= \int_0^1 w\, p(w|D)dw && \text{def of model} \\
&= \mathbf{E}[w|D] && \text{def of expectation} \\
&= \frac{|H|+a}{|H|+|T|+a+b} && \text{from Beta distribution}
\end{aligned}
$$

Thanks to the math of the Beta distribution, this nasty-looking integral simplifies down to an incredibly elegant formula for predicting the next outcome:

$p(H|D) = \frac{|H|+a}{|H|+|T|+a+b}$

Let's look back at our $\{T,T\}$ coin flip example. If we had a weak prior that the coin was fair ($a=2, b=2$), our new prediction for a Head is:

$p(H|D) = \frac{0+2}{0+2+2+2} = \frac{2}{6} = 0.33.$

Unlike MLE, which predicted 0%, our Bayesian model predicts 33%—a much more rational and safe estimate when data is scarce!

This shows what?

We can predict the class of a new sample (e.g. what will be the result of a new coin toss) based on the prior and the data.

$$p(H|D) = \frac{|H| + a}{|H| + |T| + a + b}$$

- a = b = 100: A strong prior belief that the coin is fair
- a = b = 2: A weak prior belief that the coin is fair
- a=100, b=1: For a coin biased towards heads

- If D = {T, T} , probability that next coin is H?

4. Final Summary

The lecture concludes by summarizing the two philosophical camps you now have in your toolkit:

- **The Frequentist approach (MLE):** Assumes there is one true parameter. You find it by maximizing the likelihood of your observed data. To figure out your epistemic (data) uncertainty, you have to use external computational hacks like the **Bootstrap** (resampling your data).
- **The Bayesian approach:** Treats the parameter as a random variable itself. You define a **Prior**, look at the data, and generate a **Posterior**. This posterior distribution natively quantifies your epistemic uncertainty without needing to resample anything.

Summary 1/3: Probability

- Random variables X take on values x from some domain D_X
- Probability distributions of X given by mass or density functions p
- Probabilities (masses) lie in $[0,1]$ and sum to one $\sum_{x \in D_X} p(x) = 1$
- Densities integrate to one $\int_{-\infty}^{\infty} p(x)dx = 1$
- Conditional probability $p(y|x)$
- Joint probability / product rule $p(x,y) = p(y|x)p(x)$
- Marginal probability $p(x) = \sum_{x \in D_X} p(x,y)$ or $p(x) = \int_{D_X} p(x,y)dx$
- Bayes' theorem $p(x|y) = p(y|x)p(x)/p(y)$
- Expected value $E[X]$ (if $D_X \subseteq \mathbb{R}$) $\sum_{x \in D_X} x\, p(x)$ or $\int_{D_X} x\, p(x)dx$
- Conditional expectations $E[Y \mid X = x] = \int_{D_Y} y\, p(x,y)dy$

Summary 2/3: Uncertainty

- Probabilistic models describe conditional distribution of target variable given input variables (and parameters) and thus describe aleatoric uncertainty of modelled process
- Maximum Likelihood Estimation (MLE): choose parameter values to maximise (modelled) probability of observed data
- Typically done by minimising the negative log likelihood
- Epistemic uncertainty about parameters and predictions due to finite data, i.e., could have observed different X and also different Y given X (noise)

Summary 3/3: Quantifying Uncertainty

- ❑ Bootstrap quantifies epistemic uncertainty by re-fitting model with datasets re-sampled from available data
- ❑ Results in distribution over parameters, predictions, and performance estimates
- ❑ Bayesian models instead define directly prior probability distributions on parameters
- ❑ Results in posterior distribution (given the data) of parameters and predictions that also quantifies epistemic uncertainty (use Bayes' theorem to evaluate posterior)